THANG HOANG

Department of Computer Science and Engineering, University of South Florida
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Research Interests

Applied Cryptography
Security and Privacy
Usable Security
Biometrics

Education

University of South Florida, Tampa, Florida, United States (2019 – current)
PhD Student (transferred from OSU), Computer Science
- Advisor: Dr. Attila Altay Yavuz

Oregon State University, Corvallis, Oregon, United States (2015 – 2018)
PhD Student, Computer Science
- Advisor: Dr. Attila Altay Yavuz

Chonnam National University, Gwangju, South Korea (2012 – 2014)
Master of Science, Computer Science
- Thesis: *Gait Authentication on Mobile Phone Using Pattern Recognition and Biometric Cryptosystem*
- Advisor: Dr. Deokjai Choi

University of Science VNU-HCMC, Ho Chi Minh city, Vietnam (2006 – 2010)
Bachelor of Science, Computer Science
- Thesis: *Integrating Camera Based Supervision System to Access Control Devices and Applications*
- Advisor: Dr. Thuc D. Nguyen

Professional Experiences

Research Intern, Robert Bosch Research & Technology Center, Pittsburgh, PA, United States (06.2018 – 09.2018)
- Mentor: Dr. Jorge Guajardo Merchan
- Topic: Oblivious Distributed File System

Research Intern, Robert Bosch Research & Technology Center, Pittsburgh, PA, United States (06.2016 – 09.2016)
- Mentor: Dr. Jorge Guajardo Merchan
- Topic: Distributed Searchable Symmetric Encryption

Graduate Research Assistant, Oregon State University, Corvallis, OR, United States (09.2015 – present)

Lecturer, Saigon Technology University, Ho Chi Minh city, Vietnam (09.2014 – 09.2015)
- Introduction to Programming, Java Programming with Database, Algorithms and Data Structures

Research Scientist, RedOne Technologies Co., Ltd., Gwangju, South Korea (03.2014 – 03.2015)
- WiFi Handover for Robot Platform
- Secure Communication Protocol for Robot Framework
- Security Services Supported for Robot Communication

Graduate Research Assistant, Chonnam National University, Gwangju, South Korea (03.2012 – 09.2014)
- Implicit Authentication on Mobile Phone *(funded by NRF)* (2 years)
- Mobile Device Interface Research Center (MDIRC) group (2 years)
- Wi-Fi Handover for the Robot Monitoring and Controlling System (*in collaboration with RedOne Technologies Co., Ltd.*) *(funded by NIPA)* (6 months)
- User Location Prediction *(funded by MDIRC)* (2 years)

- User Activity Recognition on Mobile Phone *(funded by MDIRC)* (1 year)

Teaching Assistant, Saigon Technology University, Ho Chi Minh city, Vietnam (09.2010 – 02.2012)

- Computer 101, Introduction to Programming, Object Oriented Programming, Web Programming

Research Assistant, Information Security Laboratory, Ho Chi Minh city, Vietnam (2009 – 02.2012)

- RFID-Based Security Platform for Access Control Management and Time Attendance Tracking Systems
- Supply Chain Management Using RFID Technologies

Publications

(Citation: 349 – H-index: 10 by Google Scholar as of 09/2019)

Journals

[1] **Thang Hoang**, Attila A. Yavuz and Jorge Guajardo, "A Multi-server ORAM Framework with Constant Client Bandwidth Blowup". *ACM Transactions on Privacy and Security* (ACM TOPS), *(to appear)*, 2019.

[2] **Thang Hoang**, Attila A. Yavuz, F. Betül Durak and Jorge Guajardo, "A Multi-Server Oblivious Dynamic Searchable Encryption Framework". *Journal of Computer Security* (JCS), IOS Press, pre-press, pp. 1-28, 2019. doi:10.3233/JCS-191300

[3] **Thang Hoang**, Attila A. Yavuz and Jorge Guajardo, "A Secure Searchable Encryption Framework for Privacy-Critical Cloud Storage Services". *IEEE Transactions on Services Computing* (IEEE TSC), in press. (SCIE). doi:10.1109/TSC.2019.2897096

[4] **Thang Hoang**, Ceyhun D. Ozkaptan, Gabriel Hackebeil and Attila A. Yavuz, "Efficient Oblivious Data Structures for Database Services on the Cloud". *IEEE Transactions on Cloud Computing* (IEEE TCC), in press. (SCI). doi:10.1109/TCC.2018.2879104

[5] **Thang Hoang**, Thuc Nguyen and Deokjai Choi, "Gait Authentication on Mobile Phone Using Biometric Cryptosystem and Fuzzy Commitment Scheme". *International Journal of Information Security* (IJIS), Volume 14, Issue 6, pp. 549–560, November 2015. (SCIE). doi:10.1007/s10207-015-0273-1

[6] **Thang Hoang** and Deokjai Choi, "Secure and Privacy Enhanced Gait Authentication on Smart Phone". *The Scientific World Journal* (TSWJ), Volume 2014, May 2014. (SCIE). doi:10.1155/2014/438254

[7] **Thang Hoang**, Thuc D. Nguyen, Chuyen Luong, Son Do and Deokjai Choi, "Adaptive Cross-Device Gait Recognition Using Mobile Accelerometer". *Journal of Information Processing System* (JIPS), Volume 9, Issue 2, pp. 333–348, June 2013. doi:10.3745/JIPS.2013.9.2.333

[8] Viet Q. Vo, **Thang Hoang** and Deokjai Choi, "Personalization in Mobile Activity Recognition System using K-Medoids Clustering Algorithm". *International Journal of Distributed Sensor Networks* (IJDSN), Volume 2013, June 2013. (SCIE). doi:10.1155/2013/315841

Conferences

[9] **Thang Hoang**, Muslum O. Ozmen, Yeongjin Jang, and Attila A. Yavuz, "Hardware-Supported ORAM in Effect: Practical Oblivious Search and Update on Very Large Dataset", in *19th Privacy Enhancing Technologies Symposium* (PETS), July 2019, Stockholm, Sweden. doi:10.2478/popets-2019-0010

[10] **Thang Hoang**, Attila A. Yavuz, F. Betül Durak, and Jorge Guajardo, "Oblivious Dynamic Searchable Encryption on Distributed Cloud Systems", in *32nd Annual IFIP WG 11.3 Conference on Data and Applications Security and Privacy* (IFIP DBSec), July 2018, Bergamo, Italy. **Best Paper Award** doi:10.1007/978-3-319-95729-6_8

[11] Muslum O. Ozmen, **Thang Hoang**, and Attila A. Yavuz, "Forward-Private Dynamic Searchable Symmetric Encryption with Efficient Search", in *IEEE International Conference on Communications* (IEEE ICC), May 2018, Kansas City, MO, USA. doi:10.1109/ICC.2018.8422480

[12] Lam Tran, **Thang Hoang**, Thuc Nguyen and Deokjai Choi, "Improving Gait Cryptosystem Security Using Gray Code Quantization and Linear Discriminant Analysis", in *International Conference on Information Security* (ISC), pp. 214–229, November 2017, Ho Chi Minh city, Vietnam.
doi:10.1007/978-3-319-69659-1_12

[13] **Thang Hoang**, Ceyhun D. Ozkaptan, Attila A. Yavuz, Jorge Guajardo and Tam Nguyen, "S^3ORAM: A Computation-Efficient and Constant Client Bandwidth Blowup ORAM with Shamir Secret Sharing", in *24th ACM Conference on Computer and Communications Security* (ACM CCS), pp. 491–505, October 2017, Dallas, Texas, United States.
doi:10.1145/3133956.3134090

[14] **Thang Hoang**, Attila A. Yavuz and Jorge Guajardo, "Practical and Secure Dynamic Searchable Encryption via Oblivious Access on Distributed Data Structure", in *32nd Annual Computer Security Applications Conference* (ACSAC), pp. 302–313, December 2016, Los Angeles, California, United States.
doi:10.1145/2991079.2991088

[15] **Thang Hoang**, Deokjai Choi and Thuc Nguyen, "On the Instability of Sensor Orientation in Gait Verification on Mobile Phone", in *12th International Conference on Security and Cryptography* (SECRYPT), pp. 148–159, July 2015, Colmar, France.
doi:10.5220/0005572001480159

[16] **Thang Hoang**, Deokjai Choi, Viet Vo, Anh Nguyen and Thuc Nguyen, "A Lightweight Gait Authentication on Mobile Phone Regardless of Installation Error", in *28th International Information Security and Privacy Conference* (IFIP SEC), pp. 83–101, July 2013, Auckland, New Zealand.
doi:10.1007/978-3-642-39218-4_7

[17] Chuyen Luong, Son Do, **Thang Hoang** and Deokjai Choi, "A Mobility Prediction Algorithm for The Seamless Handoff", in *5th International Conference Ubiquitous and Future Networks* (IEEE ICUFN), pp. 424–429, July 2013, Danang, Vietnam.
doi:10.1109/ICUFN.2013.6614854

[18] Viet Q. Vo, **Thang Hoang** and Deokjai Choi, "Adaptive Energy-Saving Strategy for Activity Recognition on Mobile Phone", in *IEEE International Symposium on Signal Processing and Information Technology* (IEEE ISSPIT), pp. 95–100, December 2012, Ho Chi Minh city, Vietnam.
doi:10.1109/ISSPIT.2012.6621267

[19] **Thang Hoang**, Viet Q. Vo, Thuc D. Nguyen and Deokjai Choi, "Gait Identification Using Accelerometer on Mobile Phone", in *1st International Conference on Control, Automation and Information Sciences* (IEEE ICCAIS), pp. 344–348, November 2012, Ho Chi Minh city, Vietnam. **Best Paper Award**
doi:10.1109/ICCAIS.2012.6466615

Domestic Publications

[20] Son Do, **Thang Hoang**, Chuyen Luong, Seungchan Choi, Dokyeong Lee, Kihyun Bang and Deokjai Choi, "Using Keystroke Dynamics for Implicit Authentication on Smartphone", in *Journal of Korea Multimedia Society*, Volumn 17, Issue 8, 2014.
doi:10.9717/kmms.2014.17.8.968

[21] **Thang Hoang** and Deokjai Choi, "A Biometric Cryptosystem Using Gait Captured from Mobile Accelerometer", in *FTRA International Symposium on Ubiquitous Computing and Embedded Systems*, December 2013, Danang, Vietnam. **Best Paper Award** (*Invited to a special issue in TSWJ [6]*)

[22] Viet Vo, **Thang Hoang**, Lee Changmoo and Deokjai Choi, "Fall Detection for Mobile Phone based on Movement Pattern". *Korean Journal of Internet Computing and Services*, Volume 13, Issue 4, pp. 23–31, 2012.
doi:10.7472/jksii.2012.13.4.23

Patents

[1] **Thang Hoang** and Jorge Guajardo, "Secure and Efficient Multi-server Oblivious Random Access Machine in a Malicious Execution Environment" *(under submission).*

[2] **Thang Hoang**, Attila A. Yavuz and Jorge Guajardo, "Practical and Secure Dynamic Searchable Encryption via Oblivious Access on Distribution Data Structure" *(under submission).*

[3] **Thang Hoang**, Muslum O. Ozmen and Attila A. Yavuz, "Forward-Private Dynamic Searchable Symmetric Encryption with Efficient Search", OSU-17-55, Provisional Application No: 62/572,339, Submitted: Oct 10, 2017.

[4] Deokjai Choi, **Thang Hoang**, Thuc D. Nguyen and Thu D. Tran, "Device and Method for Authentication System using Prime Number", Korea Patent 10-1754796, Filed: September 17, 2014, Issued: June 30, 2017.

[5] Deokjai Choi and **Thang Hoang**, "Secure Authentication System, and its Device and Method for Biometric Information, Derived Information from User Characteristic Information", Korea Patent 10-1622253, Filed: July 09, 2014, Issued: May 12, 2016.

[6] **Thang Hoang**, Deokjai Choi and Chilwoo Lee, "Gait Authentication System and its Device and Method", Korea Patent 10-1622252, Filed: December 11, 2013, Issued: May 12, 2016.

Awards and Honors

Awards and Grants

- Phi Kappa Phi Honor Society (2019)
- Student Conferenceship Travel Grant (ACSAC) (2016)
- Global IT Government (NIPA) Scholarship (2013)
- National Research Foundation of Korea Scholarship (2012–2014)
- Ministry of Science, ICT&Future Planning Research Scholarship (2012–2014)
- Brain Korea 21 (BK21 & BK21+) Research Scholarship (2012–2014)

Research Achievements

- Best Paper Award in IFIP DBSec 2018, Bergamo, Italy (2018)
- Best Paper Award in UCES 2013 international symposium, Da Nang, Vietnam (2013)
- Best Paper Award for the most innovative application in ICCAIS 2012, Ho Chi Minh city, Vietnam (2012)

Professional Services

Program Committee

- CosDEO Workshop (*affiliated with PerCom*) (2018, 2020)

Journal Reviewer

- ACM Transactions on Privacy and Security (ACM TOPS) (2017*)
- IEEE Transactions on Dependable and Secure Computing (IEEE TDSC) (2017*, 2018*)
- IEEE Transactions on Information Forensics and Security (IEEE TIFS) (2018*)
- IEEE Transactions on Cybernetics (2019*)
- IEEE Transactions on Cloud Computing (IEEE TCC) (2019)
- Elsevier Information Sciences (2017)
- Elsevier Journal of Information Security and Applications (2019)

**External reviewer*

Conference Reviewer

- ACSAC (2017, 2018, 2019)
- ASIACRYPT (2018)
- EUROCRYPT (2018)
- IFIP DBSec (2018, 2019)
- IEEE CNS (2019)
- WWW (2019)

Invited Talks

Trainer, Eduroam Workshop, University of Indonesia, Depok, Indonesia (08.2015)

- Topic: Wireless Network Security

Languages

Vietnamese	Native
English	Professional working proficiency
Korean	Elementary proficiency

Skills

C, C++, C#, VB, Matlab, Java, LaTeX, Javascript, PHP, ASP.NET